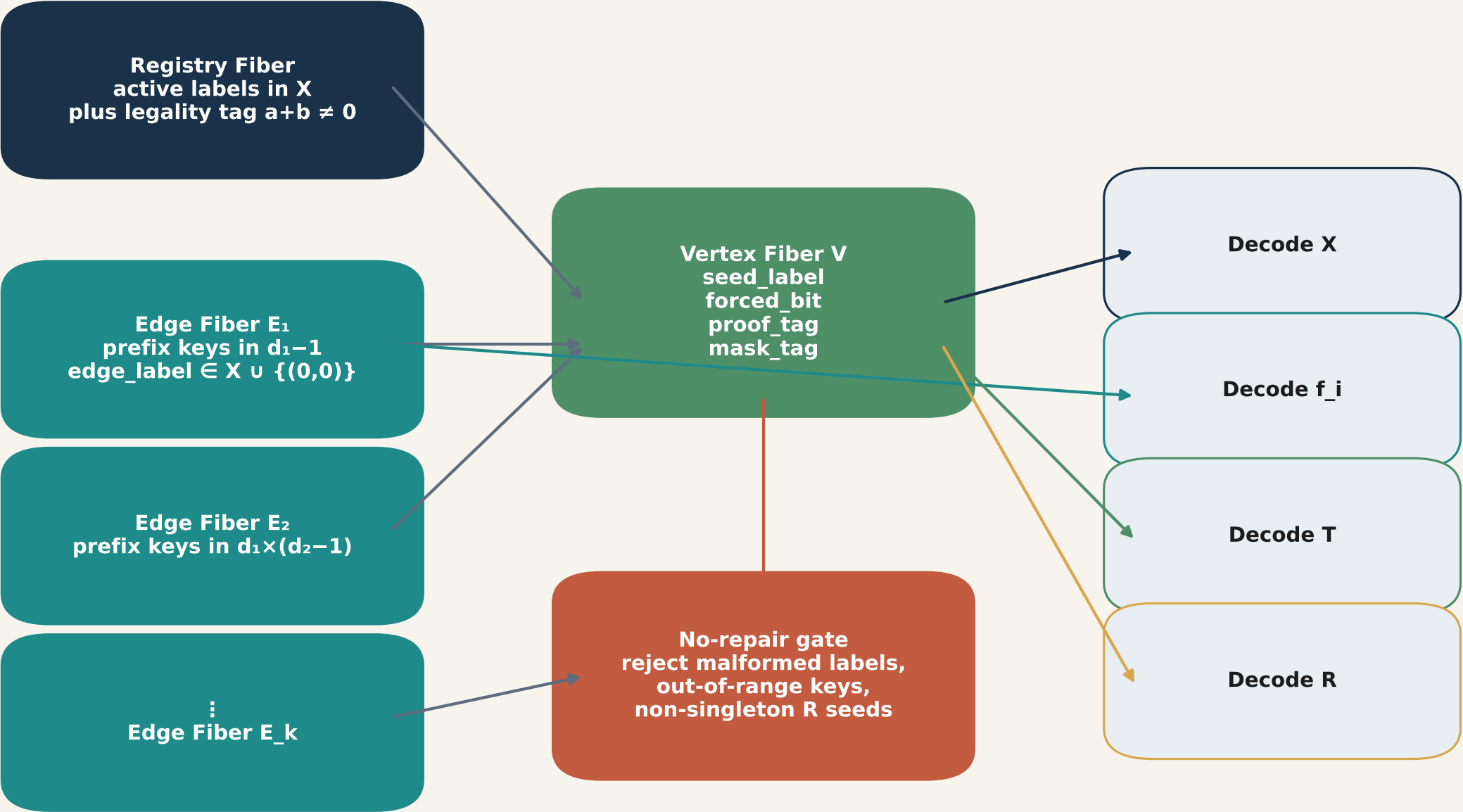


Stage-indexed proof-carrying CA used by H1



The key architectural choice is to align the automaton with the witness grammar itself: edge fibers emit partial f_i dictionaries, vertex fibers emit T and singleton-supported R seeds, and the decoder rejects rather than repairs any malformed state.